K-Nearest Neighbors (KNN)

Classification example • Jupyter Notebook-style code and output

In [1]:

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.pipeline import make_pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix
```

In [2]:

```
# Create sample classification data
X, y = make_classification(n_samples=180, n_features=2,
                           n_redundant=0, n_informative=2,
                           n_clusters_per_class=1,
                           class_sep=1.35, random_state=7)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=42, stratify=y
)
```

Out [ ]:

```
Training samples: 135
Test samples: 45
```

In [3]:

```
# Scale features and train the KNN model
model = make_pipeline(
    StandardScaler(),
    KNeighborsClassifier(n_neighbors=5)
)
model.fit(X_train, y_train)

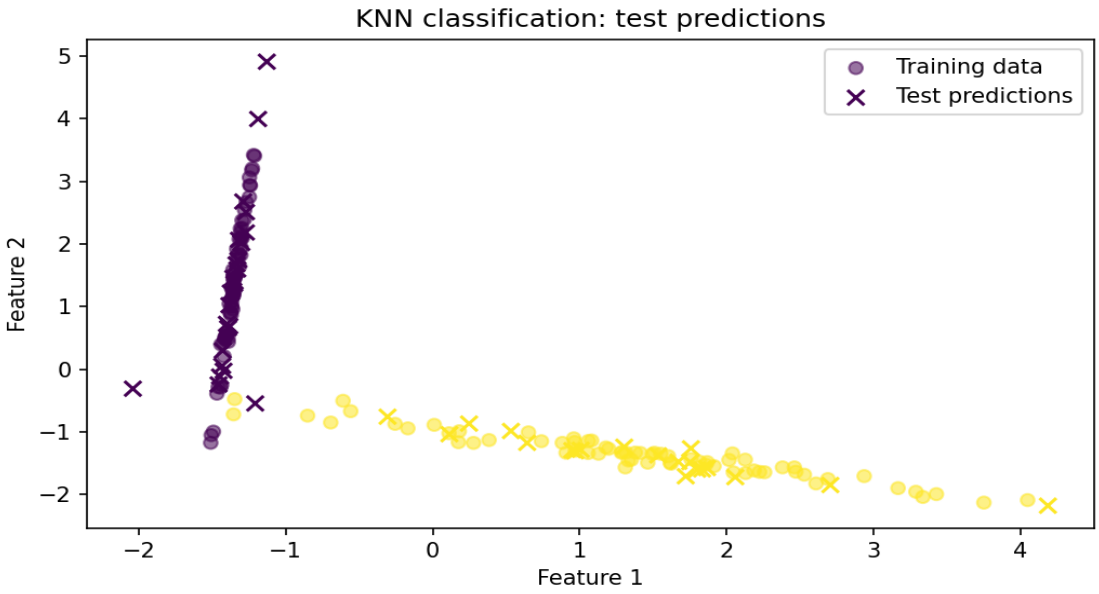
y_pred = model.predict(X_test)
print("Accuracy:", round(accuracy_score(y_test, y_pred), 3))
print("Confusion matrix:\n", confusion_matrix(y_test, y_pred))
```

Out [ ]:

```
Accuracy: 0.956
Confusion matrix:
[[23  0]
 [ 2 20]]
```

In [4]:

```
# Visualize test predictions
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train,
            alpha=0.55, label='Training data')
plt.scatter(X_test[:, 0], X_test[:, 1], c=y_pred,
            marker='x', s=55, label='Test predictions')
plt.xlabel('Feature 1')
plt.ylabel('Feature 2')
plt.title('KNN Classification: Test Predictions')
plt.legend()
plt.show()
```



Rendered output of the plotting cell